UNIT
3

CARBOHYDRATE METABOLISM



Gustav Embden, Otto Meyerhof, and Jakub Karol Parnas.

Gustav Embden (German physiological chemist), Otto Meyerhof (German physician and biochemist), and Jakub Karol Parnas (Jewish-Polish–Soviet biochemist) studied the carbohydrate metabolism and fully explained the reactions of glycolysis. They have combined results of many smaller experiments carried out by other scientists to explain the pathway as a whole. It took almost a century to fully discover the glycolytic pathway. Otto Meyerhof was awarded the Nobel Prize in Physiology and Medicine, with Archibald Vivian Hill, for his work on muscle metabolism, including glycolysis.

5UMM37

Learning Objectives

After studying this unit the students will be able to

- Describe the metabolic processes catabolism and alnabolism.
- Describe the reactions of glycolysis and gluconeogenesis.
- Describe the reactions of Hexose mono-phosphate shuttle pathway and TCA Cycle.
- Synthesis of glucose from non – carbohydrate sources
- Discuss the types, symptoms and cases of diabetes.
- Explain the protocol to determine blood glucose level.

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